

Program :

```
import ipaddress

ip = input("Enter network address: ")
prefix = int(input("Enter original prefix length: "))
subnets_required = int(input("Enter number of subnets required: "))

borrowed_bits = 0
while (2 ** borrowed_bits) < subnets_required:
    borrowed_bits += 1

new_prefix = prefix + borrowed_bits
network = ipaddress.ip_network(ip + "/" + str(prefix), strict=False)
subnets = list(network.subnets(prefixlen_diff=borrowed_bits))

print("\n===== SUBNETTING RESULT =====")
print("Original Network :", network)
print("Original Prefix :", "/" + str(prefix))
print("Required Subnets :", subnets_required)
print("Borrowed Bits  :", borrowed_bits)
print("New Prefix     :", "/" + str(new_prefix))
print("Subnet Mask    :", ipaddress.IPv4Network("0.0.0.0/" + str(new_prefix)).netmask)

hosts_per_subnet = (2 ** (32 - new_prefix)) - 2
print("Usable Hosts/Subnet :", hosts_per_subnet)

print("\n----- SUBNET DETAILS -----")

for i, subnet in enumerate(subnets[:subnets_required], 1):
    hosts = list(subnet.hosts())
```

```
print("\nSubnet", i)

print("Network Address :", subnet.network_address)

print("First Host    :", hosts[0])

print("Last Host     :", hosts[-1])

print("Broadcast     :", subnet.broadcast_address)
```

Output :

```
Output Clear
Enter network address: 192.168.1.0
Enter original prefix length: 24
Enter number of subnets required: 4

===== SUBNETTING RESULT =====
Original Network : 192.168.1.0/24
Original Prefix  : /24
Required Subnets : 4
Borrowed Bits    : 2
New Prefix       : /26
Subnet Mask      : 255.255.255.192
Usable Hosts/Subnet : 62
```

```
Output Clear
Subnet 1
Network Address : 192.168.1.0
First Host      : 192.168.1.1
Last Host       : 192.168.1.62
Broadcast       : 192.168.1.63

Subnet 2
Network Address : 192.168.1.64
First Host      : 192.168.1.65
Last Host       : 192.168.1.126
Broadcast       : 192.168.1.127

Subnet 3
Network Address : 192.168.1.128
First Host      : 192.168.1.129
Last Host       : 192.168.1.190
Broadcast       : 192.168.1.191

Subnet 4
Network Address : 192.168.1.192
First Host      : 192.168.1.193
Last Host       : 192.168.1.254
Broadcast       : 192.168.1.255
```